A multi-stage lithium-ion battery aging dataset using various experimental design methodologies

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S.1 Supplemental

S.1.1 Semi-empirical aging models

Motivated by Muehlbauer et al.[90], the total relative capacity loss of a lithium ion battery cell $Q_{\text{loss,rel}}$ is assumed to result from a linear superposition of independent calendar aging $Q^{cal}_{\text{loss,rel}}$ and cycle aging $Q^{cyc}_{\text{loss,rel}}$ related losses according

$$Q_{\text{loss,rel}} = Q^{cal}_{\text{loss,rel}} + Q^{cyc}_{\text{loss,rel}}\,, \tag{S.1}$$

where the calendar aging aspect of the model can be described by

$$Q^{\text{cal}}_{\text{loss,rel}}(T, SOC, t; \boldsymbol{\theta}_{\text{cal}}) = (1 - EOL_C) \cdot \left(e^{-\theta^{\text{cal}}_0 \cdot \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)}\right) \cdot \left(\frac{SOC}{SOC_{\text{ref}}}\right)^{\frac{1}{\theta^{\text{cal}}_1}} \cdot \left(\frac{t}{t_{\text{ref}}}\right)^{\theta^{\text{cal}}_2}\,, \tag{S.2}$$

and the cycle aging aspect by

$$\begin{aligned} &Q^{\text{cyc}}_{\text{loss,rel}}(T, SOC_{\max}, C_{\text{ch}}, C_{\text{dch}}, DOD, FEC; \boldsymbol{\theta}_{\text{cyc}}) = \\ &(1 - EOL_{\text{C}}) \cdot \left(e^{-\theta^{\text{cyc}}_0 * \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)}\right) \cdot \left(\frac{SOC_{\max}}{SOC_{\max,\text{ref}}}\right)^{\frac{1}{\theta^{\text{cyc}}_1}} \cdot \left(\frac{C_{\text{ch}}}{C_{\text{ch,ref}}}\right)^{\frac{1}{\theta^{\text{cyc}}_2}} \cdot \left(\frac{C_{\text{dch}}}{C_{\text{dch,ref}}}\right)^{\frac{1}{\theta^{\text{cyc}}_3}} \cdot \left(\frac{DOD}{DOD_{\text{ref}}}\right)^{\frac{1}{\theta^{\text{cyc}}_4}} \cdot \left(\frac{FEC}{FEC_{\text{ref}}}\right)^{\theta^{\text{cyc}}_5}. \end{aligned} \tag{S.3}$$

Both aging submodels are describing the relative capacity loss $Q_{\text{loss,rel}}$ above a capacity end-of-life criterion EOL_C, where each aging factor is scaled by a reference value and associated to one parameter of a parameter vector $\boldsymbol{\theta}$. The calendar aging model depicts the $Q^{cal}_{loss,rel}$ dependence on ambient storage temperature T and storage state of charge (SOC) during an open-loop storage condition for t days by the parameter vector $\boldsymbol{\theta}_{\text{cal}} = (\theta^{\text{cal}}_0, \theta^{\text{cal}}_1, \theta^{\text{cal}}_2)$. The cycle aging model describes the $Q^{cyc}_{loss,rel}$ dependence on ambient temperature T, maximum state of charge during cycling $SOC_{\max}$, the charge and discharge C-rates C_{ch} and C_{dch} with a cycle depth of discharge DOD along FEC fully equivalent cycles on capacity loss during cycling by its corresponding parameter vector $\boldsymbol{\theta}_{\text{cyc}} = (\theta^{\text{cyc}}_0, \cdots, \theta^{\text{cyc}}_5)$. The relative capacity extracted from the i-th reference performance

test (RPT) is always related to the initial capacity Q_0 measured during the initial RPT ("Eingangstest" - ET):

$$Q_{\mathrm{rel},i} = \frac{Q_i}{Q_0} \tag{S.4}$$

References

90. Muehlbauer, M., Rang, F., Palm, H., Bohlen, O. & Danzer, M. A. Pareto-optimal power flow control in heterogeneous battery energy storage systems. *J. Energy Storage* **48**, 103803, https://doi.org/10.1016/j.est.2021.103803 (2022).